



**RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES**

BSc (General) Degree in Applied Sciences

First Year Semester I Examination- May/June 2016

**FDN 1202 – Introduction to Computers/
CMP 1204 – Introduction to Computers**

(Theory Paper)

Time allowed: 2 hours

Library
Faculty of Applied Science
Rajarata University of Sri Lanka
Mihintale.

Instructions for candidate

- This is a closed book examination.
- The paper contains 9 pages.
- Time allowed will be 2 hours.
- The question paper consists of TWO (2) parts. **PART A** and **PART B**.
- **ANSWER ALL QUESTIONS IN PART A AND PART B.**
- This exam accounts for 70% of the subject assessment.

PART A**Answer all questions****Q1. Underline the correct answer. (10 marks)**

1. What is true about Napier's bones and logarithms?

a. Multiplication and division computer	c. It is an electro mechanical
b. Can perform addition and subtracting	d. Invented by Charles Babbage

2. Stepped Reckoner is invented by,

a. Gottfried Liebniz	c. Herman Hollerith
b. Blaise Pascal	d. Joseph-Marie Jacquard

3. Who is considered as the first computer programmer

a. Charles Babbage	c. Herman Hollerith
b. Augusta Ada Byron	d. Presper Eckert

4. Not an Electro mechanical computer

a. Mark I	c. Differential analyzer
b. Tabulating machine	d. Slide rule

5. Who found the first computer bug

a. Lady Augusta	c. Grace Hopper
b. Charles Babbage	d. Vannevar Bush

6. What is wrong about Analytical Engine?

a. Invented by Charles Babbage programming in this machine	c. Lady Ada Augusta helped in
b. Worked with punch cards	d. Use for math table calculation

7. Not a first generation computer

a. UNIVAC	c. EDSAC
b. ENIAC	d. Pascaline

8. Punch card loom was invented by

- a. Pascal
- b. Leibniz
- c. Jacquard
- d. Oughtred

9. How the data is stored in punch cards?

- a. As magnetic spots
- b. As presence of holes
- c. With the use of scales
- d. Ones and zeros

10. Pascaline was used for,

- a. Find square root
- b. For multiply only
- c. Multiply and divide
- d. Addition and subtracting

Q2. Following paragraphs describe some information regarding the computer generations. Carefully read each paragraph and fill in the blanks with suitable answers. (20 marks)

In first generation computers, Vacuum tubes are used as the equipment for information processing. They act as circuitry and memory As the memory for storing information, in this generation computers, it has been used magnetic drums There the input is made through Punch cards And Paper tapes But there were so many disadvantages in these computers such as Highly expensive and High energy consumption These machines relied on machine language which is the lowest-level programming language understood by computers, to perform operations.

In second generation computers, Technology that has been used for information processing is Transistors For the memory they used Magnetic core technology Not like in first generation computers, in this generation computers for programming they used Assembly language language. One of the example for second generation computers is Honey well 400/IBM 7030/CDC 1604

In third generation computers, they have used Integrated circuits as the technology for information processing. This device is made up of large number of transistors In these machines there were Keyboard and Magnetic tape like input devices to feed the information to computer.

In fourth generation computers, they have used Microprocessor For the processor and therefore the main advantage obtained was Smaller in size of the computers other than in the other generation computers.

We are now in fifth generation of computers. Now we can see computing devices with applications, such as voice recognition, Natural Language processing. One of the technologies they have used to build such applications is Artificial intelligence

Q3. Match the followings with suitable answers (10 marks)

- | | |
|---|---|
| 1. Computers can be classified based on (C) | A. Super computer |
| 2. Special purpose computer (E) | B. Operates on data in the form of continuously variable physical quantities. |
| 3. Digital computers (G) | C. Functionality |
| 4. Super computers (D) | D. Use for earthquake exploration |
| 5. VAX 8000 (H) | E. Has designed to handle a specific problem or a task |
| 6. Micro computer (F) | F. Mobile phone |
| 7. Mainframes (J) | G. Operated with data in numerical and digital form |
| 8. Has designed to perform range of tasks (I) | H. Mainframe Computer |
| 9. Analog computer (B) | I. General purpose computers |
| 10. IBM's Mira in United States (A) | J. Large in size |

PART B**Answer all the questions in the paper itself.**

1. State the four major functions of a computer and specify the basic component which is responsible for each function. (8 marks)

1.Input- Keyboard,Mouse

2.Processing-CPU

3.Output-Monitor,Printer

4.Storage-Primary memory(RAM,ROM),Secondary memory(CD,DVD)

2. Depict the memory hierarchy and state what happens when going downwards in the hierarchy and when going upwards in the hierarchy. (8 marks)

3. Solve $ABE5_{16} + 356_8$ and keep the answer in binary form (8 marks)

A= 10	8	4	2	1		4	2	1
B=11	A= 1	0	1	0		3 = 0	1	1
C=12	B= 1	0	1	1		5 = 1	0	1
D=13	E= 1	1	1	0		6 = 1	1	0
	5= 0	1	0	1				
ABE5	= 1010 1011 1110 0101					356	= 011 101 110	
16						8		
ABE5	=	1010101111100101						
356	=	011101110						
8		1010110011010011						

4. Solve using 2's complement $35 - 12$ (8 marks)

	128	64	32	16	8	4	2	1
35	=	0	0	1	0	0	0	1
12	=	0	0	0	0	1	1	0
-12(1's)	=	1	1	1	1	0	0	1
-12(2's)	=							1
		1	1	1	1	0	1	0

35	=	0	0	1	0	0	0	1
-12(2's)	=	1	1	1	1	0	1	0
		1	0	0	0	1	1	1

Carry bit(In 2's Complement we neglect the carry bit)

128	64	32	16	8	4	2	1
0	0	0	1	0	1	1	1

$$16 + 4 + 2 + 1 = 23$$

5. Mention the difference between compiler and interpreter. (4 marks)

6. Write 4 functions done by Input Output module. (4 marks)

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7. What are the functions of an Operating system in a computer? (8 marks)

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8. Under the law of copyright, mention 3 things you are prohibited to do for a copyrighted material? (3 marks)

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9. Explain what software piracy is. (3 marks)

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